

Addenda.

I.

Irreducibility of the cyclotomic equation.

The last part of §. 134 of the first volume, which treats the irreducibility of the cyclotomic equation in the general case in which the degree of the root of unity is a number composed of several different prime numbers, requires a correction, which is to be supplied here; and we ask the reader to complete the cited passage according to what is here carried out.

If one puts $n = ab$, with a and b relatively prime, and if ρ, α, β are primitive n th, a th, b th roots of unity, then $\rho = \alpha\beta$. At the cited place it was proved that, if ρ is a root of a rational equation $\Phi(x) = 0$, then among the roots of this equation $\rho = \alpha\beta$ all primitive a th roots of unity α and all primitive b th roots of unity β must occur. But it does not follow from this, as was erroneously assumed there, that among them *all* primitive n th roots of unity occur, since it was not shown that every α occurs combined with every β .

To fill this gap, there may first be given here a wholly elementary proof¹.

If p is a prime number, then all binomial coefficients for the p th power

$$B_h^{(p)} = \frac{p(p-1) \cdots (p-h+1)}{1 \cdot 2 \cdots h},$$

with the exception of $B_0^{(p)}$ and $B_p^{(p)}$, are integers divisible by p , and consequently, if u, v are indeterminate quantities and the congruence is referred only to the coefficients,

$$(u+v)^p \equiv u^p + v^p \pmod{p}.$$

If in this formula we replace v again by a sum $v + w + \cdots$, we obtain, by complete induction (or also from the polynomial theorem),

$$(u+v+w+\cdots)^p \equiv u^p + v^p + w^p + \cdots \pmod{p},$$

and, if we apply this formula repeatedly and understand by k an arbitrary positive exponent,

$$(u+v+w+\cdots)^{p^k} \equiv u^{p^k} + v^{p^k} + w^{p^k} + \cdots \pmod{p}. \quad (1)$$

Let x now be a variable, and let $a_1, a_2, \dots, a_\nu$ be rational integers,

$$f(x) = x^\nu + a_1 x^{\nu-1} + a_2 x^{\nu-2} + \cdots + a_\nu$$

a whole function of x whose roots may be $\alpha, \beta, \gamma, \dots$, so that

$$-a_1 = \sum \alpha, \quad a_2 = \sum \alpha\beta, \quad -a_3 = \sum \alpha\beta\gamma \cdots$$

are the elementary symmetric functions of $\alpha, \beta, \gamma, \dots$

We form the function

$$F(x) = x^\nu + A_1 x^{\nu-1} + A_2 x^{\nu-2} + \cdots + A_\nu,$$

whose roots are the p^k th powers of the roots of f , thus

$$\alpha^{p^k}, \beta^{p^k}, \gamma^{p^k}, \dots$$

¹Arndt, Crelle's Journal, vol. 56, p. 178 (1859). Lebesgue, Liouville's Journal, 2nd series, vol. 4, p. 105 (1859).

The coefficients of F ,

$$A_1 = -\sum \alpha^{p^k}, \quad A_2 = \sum \alpha^{p^k} \beta^{p^k}, \quad A_3 = -\sum \alpha^{p^k} \beta^{p^k} \gamma^{p^k} \dots,$$

are, as integral symmetric functions of $\alpha, \beta, \gamma, \dots$, rationally and integrally expressible by $a_1, a_2, \dots$, and hence are integers. The quotients

$$\frac{A_1 - a_1^{p^k}}{p}, \quad \frac{A_2 - a_2^{p^k}}{p}, \quad \frac{A_3 - a_3^{p^k}}{p}, \dots$$

are likewise integers by formula (1), and hence by Fermat's theorem

$$F(x) \equiv f(x) \pmod{p}. \quad (2)$$

These propositions are now to be applied to the function $f_n(x)$ whose roots are all primitive n th roots of unity $\alpha, \beta, \gamma, \dots$, and which, as was proved in §. 133 of the first volume, has integral coefficients $a_1, a_2, \dots$

If p is a prime number dividing n , then

$$\frac{x^n - 1}{x^{\frac{n}{p}} - 1} = x^{\frac{n}{p}(p-1)} + x^{\frac{n}{p}(p-2)} + \dots + x^{\frac{n}{p}} + 1, \quad (3)$$

and this function vanishes for $x = \alpha, \beta, \gamma, \dots$, and is therefore divisible by $f_n(x)$. We shall put it equal to $f_n(x)g(x)$, where $g(x)$ is then (by vol. I, §. 2) a whole function of x with integral coefficients.

If now

$$n = p^k n',$$

where n' is no longer divisible by p , and if α' is a primitive n' th root of unity, then

$$\alpha'^{\frac{n}{p}} = 1,$$

and consequently, if one puts $x = \alpha'$ [by (3)],

$$f_n(\alpha')g(\alpha') = p. \quad (4)$$

From this we first derive a proof of the irreducibility of $f_n(x)$ under the assumption that n is a power of a prime, although for this case the proof given in vol. I, §. 134 is unobjectionable.

Assume that $f_n(x)$ decomposes into two rational factors:

$$f_n(x) = \varphi(x)\psi(x). \quad (5)$$

Then all roots, both of $\varphi(x)$ and of $\psi(x)$, are n th roots of unity, and their n th powers are therefore equal to 1. If now $n = p^k$, then from $\varphi(x)$ and $\psi(x)$ we can derive two whole functions $\Phi(x)$ and $\Psi(x)$ whose roots are the n th powers of the roots of $\varphi(x)$ and $\psi(x)$, and therefore are all equal to 1; and from (2) one obtains

$$\varphi(x) \equiv \Phi(x), \quad \psi(x) \equiv \Psi(x) \pmod{p},$$

whence for $x = 1$ it follows that

$$\varphi(1) \equiv 0, \quad \psi(1) \equiv 0 \pmod{p}.$$

From this (5) gives

$$f_n(1) \equiv 0 \pmod{p^2}. \quad (6)$$

But in the present case $f_n(1) = p$ (vol. I, §. 133, V.), which contradicts formula (6). Our assumption was therefore inadmissible, and $f_n(x)$ is irreducible.

To prove the irreducibility for an arbitrary composite n , we apply complete induction. We put

$$n = p^k n',$$

and understand by p a prime number which divides n but not n' , and assume the irreducibility of $f_{n'}(x)$ as already proved. If $f_n(x)$ is then decomposable into two rational factors $\varphi(x)$, $\psi(x)$, neither of which is constant,

$$f_n(x) = \varphi(x)\psi(x), \quad (7)$$

then from $\varphi(x)$ we derive the function $\Phi(x)$ whose roots are the p^k th powers of the roots of $\varphi(x)$, and which satisfies the condition

$$\varphi(x) \equiv \Phi(x) \pmod{p},$$

or, if we denote by $\chi(x)$ a second integral function of x ,

$$\varphi(x) = \Phi(x) + p\chi(x).$$

The roots of $\Phi(x)$, however, are primitive roots of unity of degree n' , and therefore among them there must be at least one, α' , for which $\Phi(\alpha')$ vanishes, so that

$$\varphi(\alpha') = p\chi(\alpha').$$

Because of the assumed irreducibility of $f_{n'}(x)$, however, this equation must hold for *all* roots of unity α' ; and for the same reason, if $\vartheta(x)$ is an integral function, for all α' one has

$$\psi(\alpha') = p\vartheta(\alpha').$$

Accordingly (7) gives

$$f_n(\alpha') = p^2\chi(\alpha')\vartheta(\alpha'),$$

and by (4),

$$1 = p\chi(\alpha')\vartheta(\alpha')g(\alpha').$$

But the product $\chi(x)\vartheta(x)g(x)$ is a whole integral function of x , whose remainder on division by $f_{n'}(x)$ likewise has integral coefficients and may be denoted by $\Theta(x)$. This remainder is of lower degree than $f_{n'}(x)$, and the function

$$p\Theta(x) - 1$$

vanishes for all roots of $f_{n'}(x)$. Consequently the equation

$$p\Theta(x) = 1$$

must be identical, which is plainly impossible, since $\Theta(x)$, for an integral x , passes into an integer which, multiplied by p , cannot be equal to 1.

Thus the irreducibility of $f_n(x)$ is proved in general.

After the irreducibility of the cyclotomic equation $f_n(x) = 0$ has thus been proved in general, it follows, as in §. 134 of the first volume, that it remains irreducible when an arbitrary root of unity is adjoined whose degree is relatively prime to n .

Kronecker goes, in a certain sense, the opposite way². He first proves, under the assumption that n is a power of a prime p , the irreducibility of $f_n(x)$ after adjunction of a root of unity

²Mém. sur les facteurs irréduct. de l'expression $(x^n - 1)$. Liouville's Journal, vol. 19 (1854). A simple proof of the irreducibility of the general cyclotomic equation based on the theory of higher congruences was given by Dedekind [Crelle's Journal für Mathematik, vol. 58 (1859)].

whose degree is not divisible by p , and derives from this the general proof of irreducibility, as we shall now briefly show.

Let now

$$n = p^k$$

be a power of a prime number p , and let

$$f_n(x) = x^{p^{k-1}(p-1)} + x^{p^{k-1}(p-2)} + \dots + x^{p^{k-1}} + 1 \quad (8)$$

be the function whose roots are the primitive n th roots of unity. Further let α be any root of unity whose degree m is not divisible by p , and let ν be the degree of the field $R(\alpha)$. It is to be proved that $f_n(x)$ is irreducible in the field $R(\alpha)$.

Every number of the field $R(\alpha)$ can be represented in one and only one way in the form

$$\varphi(\alpha) = m_0 + m_1\alpha + \dots + m_{\nu-1}\alpha^{\nu-1}, \quad (9)$$

where $m_0, m_1, \dots, m_{\nu-1}$ are rational numerical coefficients.

If $\chi(x) = 0$ is the rational equation of least degree whose root is α , and

$$\chi(x) = x^\nu + c_1x^{\nu-1} + \dots + c_{\nu-1}x + c_\nu, \quad (10)$$

then $c_1, c_2, \dots$ are integers, since $\chi(x)$ must be a divisor of $x^m - 1$ (by Gauss's theorem, vol. I, §. 2).

From this it follows that, if $\varphi(x)$ denotes a whole function of x with integral rational coefficients, then, even if the degree of $\varphi(x)$ was originally higher than ν , the coefficients $m_0, m_1, \dots, m_{\nu-1}$ in (9) become integers; for these coefficients are obtained when one seeks the remainder of the division of $\varphi(x)$ by $\chi(x)$.

We now assume that $f_n(x)$ is decomposable in the field $R(\alpha)$, and thus that

$$f_n(x) = \varphi(x, \alpha)\psi(x, \alpha), \quad (11)$$

where $\varphi(x, \alpha)$ and $\psi(x, \alpha)$ are two whole functions of x , neither of which is constant, and whose coefficients are contained in $R(\alpha)$. From (8) and (11), if we put $x = 1$, we obtain

$$p = \varphi(1, \alpha)\psi(1, \alpha), \quad (12)$$

where $\varphi(1, \alpha)$ and $\psi(1, \alpha)$ are representable, as numbers of the field $R(\alpha)$, in the form (9). If we denote the least common denominators of these two representations by a and b , we get

$$\begin{aligned} a\varphi(1, \alpha) &= a_0 + a_1\alpha + a_2\alpha^2 + \dots + a_{\nu-1}\alpha^{\nu-1} = A(\alpha), \\ b\psi(1, \alpha) &= b_0 + b_1\alpha + b_2\alpha^2 + \dots + b_{\nu-1}\alpha^{\nu-1} = B(\alpha), \end{aligned} \quad (13)$$

where $a, a_0, a_1, \dots, a_{\nu-1}$ are integers without a common divisor, and A is a function-symbol. The same holds for $b, b_0, b_1, \dots, b_{\nu-1}$ and B .

The roots of the equation $\varphi(x, \alpha) = 0$ are found among the primitive n th roots of unity, and these are all powers of one among them, r . Accordingly there are certain powers $r^h, r^{h'}, r^{h''}, \dots$ such that

$$\varphi(x, \alpha) = (x - r^h)(x - r^{h'})(x - r^{h''}) \dots,$$

and hence, by (13),

$$A(\alpha) = a(1 - r^h)(1 - r^{h'})(1 - r^{h''}) \dots$$

If we take the n th power of this, then, by (1), taking into account the equation $r^n = 1$, we obtain

$$[A(\alpha)]^n = p\chi(r),$$

where $\chi(x)$ denotes a whole function of the variable x with integral coefficients.

Now, understanding by t a variable, we set

$$[t - \chi(1)][t - \chi(r)] \cdots [t - \chi(r^{n-1})] = t^n + A_1 t^{n-1} + A_2 t^{n-2} + \cdots + A_n,$$

where the coefficients $A_1, A_2, \dots, A_n$, as whole symmetric functions of the n quantities $\chi(1), \chi(r), \dots, \chi(r^{n-1})$, are rational integers. But if in this equation we put

$$t = \chi(r) = \frac{[A(\alpha)]^n}{p},$$

then the left hand side vanishes, and multiplication by p^n gives

$$[A(\alpha)]^{n^2} = pC(\alpha), \quad (14)$$

where $C(\alpha)$ denotes a number of the form (9) with integral coefficients.

If formula (14) is repeatedly raised to the p th power and proposition (1) is applied, then for every power not smaller than n^2 one obtains

$$A(\alpha^{p^\lambda}) = p\xi(\alpha), \quad (15)$$

where the whole function $\xi(\alpha)$ likewise has integral coefficients.

Here we may assume λ divisible by $\varphi(m)$; then, since m is not divisible by p , by the generalized theorem of Fermat (vol. II, §. 14) one has

$$p^\lambda \equiv 1 \pmod{m}, \quad \alpha^{p^\lambda} = \alpha,$$

so that from (14) it follows that

$$A(\alpha) = p\xi(\alpha). \quad (16)$$

Thus in (13) the coefficients $a_0, a_1, \dots, a_{\nu-1}$ are all divisible by p , and consequently a is not divisible by p . In the same way it is shown that $b_0, b_1, \dots, b_{\nu-1}$ are divisible by p , while b is indivisible by p .

Consequently from (12), if one puts $A(\alpha)B(\alpha) = p^2\Theta(\alpha)$, one obtains

$$ab = p\Theta(\alpha), \quad (17)$$

where $\Theta(\alpha)$ denotes a function of α of the form (9) with integral coefficients. But the equation (17), since it is of at most the $(\nu - 1)$ st degree, would have to be identical with respect to α , and ab would have to be divisible by p , which is impossible.

It follows from this that our assumption that $f_n(x)$ is decomposable in the field $R(\alpha)$ into two factors according to formula (11) is inadmissible.

This now again gives a general proof that the equation whose roots are all primitive m th roots of unity, and whose degree is $\varphi(m)$, is irreducible even when m is an arbitrary composite number. This is equivalent to the following proposition:

If a whole function $\varphi(x)$ with rational coefficients vanishes when a primitive m th root of unity is substituted for x , then $\varphi(x)$ also vanishes when any other primitive m th root of unity is substituted for x .

If m is a power of a prime, then the proposition has already been proved above. We therefore apply complete induction, assume it proved for m th roots of unity, and derive it for m nth roots of unity, where $n = p^k$ and p is a prime not dividing m .

Every m nth root of unity β can be represented in the form

$$\beta = \alpha r,$$

where α is an m th and r an n th root of unity. What is to be proved can be expressed as follows: if β is the root of a rational equation $\Phi(x) = 1$, then all other primitive roots of unity of degree mn must also be roots of this equation.

If, as above, $f_n(x) = 0$ is the equation whose roots are the quantities r , then, as we have proved, $f_n(x)$ is not decomposable into rational factors, nor into such factors as depend rationally on α . The function $f_n(\alpha^{-1}x)$, which vanishes for $x = \beta$, likewise cannot be decomposed into rational factors; for from a decomposition of the form

$$f_n(\alpha^{-1}x) = \varphi(x, \alpha)\psi(x, \alpha)$$

one would obtain

$$f_n(x) = \varphi(\alpha x, \alpha)\psi(\alpha x, \alpha),$$

which cannot hold. Since accordingly $f_n(\alpha^{-1}x)$ has a root in common with $\Phi(x)$, $\Phi(x)$ must be divisible by $f_n(\alpha^{-1}x)$, and we can therefore, for an indeterminate x , put

$$\Phi(x) = f_n(\alpha^{-1}x)\Psi(x, \alpha), \quad (18)$$

where $\Psi(x, \alpha)$ is a whole rational function of x .

Let $\alpha, \alpha_1, \alpha_2, \dots$ be all the primitive m th roots of unity, which, as we have assumed, are roots of an irreducible rational equation

$$F_m(x) = 0.$$

Then, by this assumption, in the identity (18) α can be replaced by every other $\alpha_1, \alpha_2, \dots$, and it follows that $\Phi(x)$ is divisible by each of the factors

$$f_n(\alpha^{-1}x), \quad f_n(\alpha_1^{-1}x), \quad f_n(\alpha_2^{-1}x) \cdots$$

But no two of these factors can have a common divisor; for if, for instance,

$$f_n(\alpha^{-1}x) = 0, \quad f_n(\alpha_1^{-1}x) = 0$$

had a common root x , then there would have to exist two primitive n th roots of unity r, r_1 such that

$$x = \alpha r = \alpha_1 r_1.$$

Thus one would have $\alpha^n = \alpha_1^n$, and consequently, since n is relatively prime to m , $\alpha = \alpha_1$.

Accordingly $\Phi(x)$ is divisible by the product

$$F_m^{(n)}(x) = f_n(\alpha^{-1}x)f_n(\alpha_1^{-1}x)f_n(\alpha_2^{-1}x) \cdots,$$

whose roots are all the β ; and therefore this function is also indecomposable, as was to be proved.

II.

The irreducibility of the cyclotomic equation
and the theorem on the prime numbers contained in a linear form.

Whereas in the first addendum, which is to be regarded as a supplement to the first volume, no use was deliberately made of the theory of whole algebraic numbers, we give in conclusion a proof of the irreducibility of the cyclotomic equation that rests on other foundations, namely on

the theorems on the class number (§. 192), and that is noteworthy because of the generalizations which it permits.

Let m be any natural number, and let n be the system of positive numbers prime to m ; among these there are

$$\varphi(m) = \mu \quad (1)$$

which are smaller than m ; we denote them by a .

If we unite all numbers n congruent modulo m to one such a into a class A , we obtain μ number-classes

$$A_1, A_2, \dots, A_\mu, \quad (2)$$

which, under composition by multiplication, form an Abelian group $\mathfrak{A}$ of degree μ . This group was already considered in §. 16 of this volume.

The μ characters of this group we denote by

$$\chi_1, \chi_2, \dots, \chi_\mu, \quad (3)$$

and put, if χ is one of these characters and n denotes a number from the class A ,

$$\chi(A) = \chi(n). \quad (4)$$

These characters, which were determined more closely in §. 16, are all μ th roots of unity.

Here, however, we do not need to make use of the special expressions of these characters. Rather, we rely on the following theorem, which holds for all Abelian groups:

1. If A is an element of f th degree of an Abelian group of μ th degree, and if $\mu = ef$, then all $\chi_i(A)$ are f th roots of unity, and among them each f th root of unity occurs exactly e times.

The theorem is contained implicitly in theorem 7, §. 12 of this volume. For if we apply this theorem to the group

$$T = 1, A, A^2, \dots, A^{f-1},$$

whose index with respect to $\mathfrak{A}$ is equal to e , then it follows that there is a group $\mathfrak{F}$ of e characters ξ which satisfy the conditions

$$\xi(A) = 1.$$

If, therefore, the group X of the characters χ is decomposed into the cosets

$$\mathfrak{F}_1, \mathfrak{F}_2, \dots, \mathfrak{F}_f,$$

then $\chi_1(A)$ and $\chi_2(A)$ are equal to one another or different from one another according as χ_1 and χ_2 occur in the same or in different ones of these cosets. Moreover, since, on account of $\chi(A)^f = \chi(A^f) = 1$, all $\chi(A)$ are roots of unity of degree f , and there are only f different such roots, our theorem is thereby proved.

If n is contained in the class A , then the degree f of A is the exponent to which n belongs modulo m , that is, the least positive exponent for which

$$n^f \equiv 1 \pmod{m} \quad (5)$$

holds. If therefore n belongs to the exponent f , then all $\chi_i(n)$ are f th roots of unity, and each f th root of unity occurs among them e times.

Let a denote any one of the $\varphi(m)$ residues of the numbers n . Then, in the form

$$\mu_x = mx + a - m \quad (6)$$

every number of the number-class (2) represented by a is contained; if by a we understand the least positive residue, then we obtain the totality of the numbers of this class by letting x run through all positive integral values.

Now

$$\frac{\mu_x}{m} - x = \frac{a - m}{m},$$

and therefore we can apply the theorem §. 191, 6. to the quantities μ_x , where we have to put $n = x$, $\gamma = 1 : m$, $c_n = (a - m) : m$. Thus

$$A(s) = \sum_{x=0}^{\infty} \frac{1}{(mx + a)^s} = \sum_{x=1}^{\infty} \frac{1}{(mx + a - m)^s}, \quad (7)$$

as long as $s > 1$, is a continuous function of s , and

$$A(s) = \frac{1}{m(s-1)} + C(s), \quad (8)$$

where $C(s)$ remains finite for $s = 1$ ³.

Now we let A run through all classes (2), denote by χ any one of the characters of the group $\mathfrak{R}$, and put

$$Q(s) = \sum_A \chi(A) A(s), \quad (9)$$

or, what is the same,

$$Q(s) = \sum_n \frac{\chi(n)}{n^s}, \quad (10)$$

where the sum extends over all numbers n . The number of these sums Q is as large as the number of characters, that is, $= \varphi(m)$. If a distinction is necessary, we denote them by

$$Q_1, Q_2, \dots, Q_\mu.$$

One of them, Q_1 , corresponds to the principal character and has the expression

$$Q_1(s) = \sum_n \frac{1}{n^s}.$$

Now [§. 11, (21)]

$$\sum_A \chi(A) = \mu \quad \text{or} = 0,$$

according as χ is the principal character or not. Furthermore, from (8) one obtains

$$Q(s) = \frac{1}{m(s-1)} \sum_A \chi(A) + \sum_A \chi(A) C(s),$$

and hence the theorem:

2. The sums $Q_2, \dots, Q_\mu$ receive finite values for $s = 1$; Q_1 becomes infinite, and indeed

$$\lim_{s \rightarrow 1} (s-1) Q_1(s) = \frac{\mu}{m}.$$

³By using known theorems on Γ -functions, one finds from §. 191, (20):

$$C(1) = \frac{1}{m} \frac{\Gamma'(\frac{a}{m})}{\Gamma(\frac{a}{m})}.$$

To transform the sums $Q(s)$, we denote by p each prime number not dividing m and multiply together all infinite series of the form

$$1 + \frac{\chi(p)}{p^s} + \frac{\chi(p^2)}{p^{2s}} + \cdots = \frac{1}{1 - \chi(p)p^{-s}}.$$

In this way we obtain [cf. §. 192, (8)]

$$Q(s) = \prod_p \frac{1}{1 - \chi(p)p^{-s}}. \quad (11)$$

If now p belongs to the exponent f , so that f is the least positive number satisfying the congruence

$$p^f \equiv 1 \pmod{m}, \quad (12)$$

and if $\mu = ef$, then among the $\chi(p)$ each f th root of unity occurs exactly e times; consequently, for a variable x ,

$$\prod_{\chi} [1 - \chi(p)x] = (1 - x^f)^e,$$

and therefore from (11) one obtains

$$Q_1 Q_2 \cdots Q_{\mu} = \prod_p \frac{1}{(1 - p^{-fs})^e}. \quad (13)$$

We now introduce the cyclotomic field Ω_m , without presupposing the irreducibility of the cyclotomic equation, and denote its still unknown degree by ν . Since, however, the primitive m th root of unity r satisfies a rational equation of degree μ , one has

$$\nu \leq \mu. \quad (14)$$

The numbers ω of the field Ω_m are all representable in the form

$$a\omega = a_0 + a_1 r + \cdots + a_{\nu-1} r^{\nu-1} \quad (15)$$

with rational integer coefficients $a, a_0, a_1, \dots, a_{\nu-1}$, and if ω is a whole number, then for a we can put a fixed integer, whose closer knowledge is not required. Its value is equal to the square root of the quotient of the discriminants

$$\Delta(1, r, r^2, \dots, r^{\nu-1}) : \Delta(\omega_1, \omega_2, \dots, \omega_{\nu}),$$

where $\omega_1, \omega_2, \dots, \omega_{\nu}$ is a minimal basis (§. 144, 145).

We exclude from the system of the p all prime numbers occurring in the discriminant of the cyclotomic equation and all occurring in m and in a ; their number is in any case finite. Let $\mathfrak{p}$ denote a prime ideal occurring in p . Then the congruence

$$r^p \equiv r \pmod{\mathfrak{p}}$$

is fulfilled precisely when

$$p \equiv 1 \pmod{m}, \quad (16)$$

that is, when p belongs to the exponent 1, hence when $r^p = r$; under the same condition, by (15), one also has, for every whole number ω in Ω_m ,

$$\omega^p \equiv \omega \pmod{\mathfrak{p}}.$$

But by §. 162 this is the necessary and sufficient condition that the prime number p , not occurring in the field-discriminant of Ω_m , decomposes into prime ideals of the first degree only.

Thus, if we denote by

$$P_f(s) = \prod (1 - p^{-sf})$$

the product extending over all prime numbers p belonging to the exponent f , it follows from §. 192 that, if $f > 1$, $P_f(1)$ is finite and different from zero, and that

$$\frac{s-1}{P_1(s)^\nu} \quad (17)$$

for $s = 1$ is finite and different from zero.

Now, by (13), if S denotes the finite product $\prod (1 - p^{-sf})^{-e}$ extending over the excluded prime numbers p ,

$$Q_1 Q_2 \cdots Q_\mu = S \prod_f \frac{1}{P_f(s)^e},$$

and, since $e = \mu$ for $f = 1$, it follows from (17) that

$$(s-1)^{\frac{\mu}{\nu}} Q_1 Q_2 \cdots Q_\mu \quad (18)$$

for $s = 1$ is finite and different from zero.

On the other hand, by 2.,

$$(s-1) Q_1 Q_2 \cdots Q_\mu \quad (19)$$

is finite for $s = 1$, and if we therefore divide (19) by (18), it follows that

$$(s-1)^{1-\frac{\mu}{\nu}} \text{ is finite for } s = 1,$$

that is,

$$\nu \geq \mu. \quad (20)$$

Together with (14), this gives $\nu = \mu$, whereby the following theorem is proved:

3. The degree of the cyclotomic field Ω_m is equal to $\varphi(m)$; hence the cyclotomic equation of degree $\varphi(m)$, whose roots are the primitive m th roots of unity, is irreducible.

This consideration, however, also renders us another important service, by furnishing a proof of the famous theorem that *in every arithmetic progression whose difference and initial term are natural numbers without common divisor, infinitely many prime numbers are contained*⁴.

For it now follows from (18) that the product

$$(s-1) Q_1 Q_2 \cdots Q_\mu$$

has, for $s = 1$, a finite and non-zero value; and since by 2. none of the factors is infinite, it follows that

4. The sums

$$(s-1) Q_1(s), \quad Q_2(s), \dots, Q_\mu(s)$$

have finite and non-zero values for $s = 1$.

⁴The first complete and general proof of this theorem was given by Dirichlet (Abhandlungen der Berliner Akademie for the year 1837; collected works no. XXI). Dedekind drew attention to the essential simplification of this proof by use of Kummer's formulas for the class number in cyclotomic fields (Vorlesungen über Zahlentheorie, 3rd edition, p. 596).

If we apply to all factors of (11) the formula

$$\log[1 - \chi(p)p^{-s}] = \frac{\chi(p)}{p^s} + \frac{1}{2} \frac{\chi(p^2)}{p^{2s}} + \frac{1}{3} \frac{\chi(p^3)}{p^{3s}} + \dots$$

(where the imaginary part of the logarithm is to be taken between $\pm\pi i$), then, if the imaginary part of $\log Q(s)$ is suitably determined, it follows that

$$\log Q(s) = \sum \frac{\chi(p)}{p^s} + \frac{1}{2} \sum \frac{\chi(p^2)}{p^{2s}} + \frac{1}{3} \sum \frac{\chi(p^3)}{p^{3s}} + \dots \quad (21)$$

If now A denotes any one of the classes (2), and a a number from A , then by §. 11, (6) of this volume

$$\sum_{\chi} \chi(A) \chi(n) = \sum_{\chi} \chi(an) = \mu \quad \text{or} = 0, \quad (22)$$

according as $an \equiv 1$ or not $\equiv 1$ modulo m , that is, according as n is contained in the class A^{-1} or not. If, therefore, we multiply formula (21) by $\chi(A)$ and sum with respect to χ , then it follows that

$$\frac{1}{\mu} \sum_{\chi} \chi(A) \log Q(s) = \sum \frac{1}{p^s} + \frac{1}{2} \sum \frac{1}{p^{2s}} + \frac{1}{3} \sum \frac{1}{p^{3s}} + \dots, \quad (23)$$

where on the right the first, second, third, ... sum refers to all prime numbers p whose first, second, third, ... power is contained in A^{-1} .

The second part of this sum,

$$R(s) = \frac{1}{2} \sum \frac{1}{p^{2s}} + \frac{1}{3} \sum \frac{1}{p^{3s}} + \dots,$$

is enlarged if each of its partial sums is extended over *all* prime numbers p , and consequently

$$R(s) < \frac{1}{2} \sum \left(\frac{1}{p^{2s}} + \frac{1}{p^{3s}} + \dots \right) = \frac{1}{2} \sum \frac{1}{p^s(p^s - 1)},$$

and therefore still more

$$R(s) < \frac{1}{2} \sum_{n=2}^{\infty} \frac{1}{n^{2s}} \frac{1}{1 - n^{-s}},$$

or, since $1 - n^{-s} > \frac{1}{2}$,

$$R(s) < \sum_{n=2}^{\infty} \frac{1}{n^{2s}}.$$

From this one concludes that R remains finite for $s = 1$ (§. 191, 2.).

If now in formula (23),

$$\frac{1}{\mu} \sum_{\chi} \chi(A) \log Q(s) = \sum \frac{1}{p^s} + R(s),$$

we let s pass to 1, then $\log Q_1(s)$ becomes infinite by 4., while the remaining terms of the left hand side, $\log Q_2(s), \dots, \log Q_{\mu}(s)$, remain finite. Since $R(s)$ likewise remains finite, the sum $\sum p^{-s}$ must become infinite; and this is certainly possible only if the sum consists of infinitely many terms. Hence it is proved that:

5. In each of the number-classes A (modulo m) infinitely many prime numbers are contained.

This can also be expressed by saying that the linear form $mx + a$, in which m and a are integers without a common divisor, represents a prime number for infinitely many integral values of x .